

TOAE209/TOAH209 – Design and Analysis of Algorithms

Week 11: Theoretical Questions

Foreign Trade University

Instructions

Part A contains 17 questions requiring written explanations or short proofs (including the key proofs of Hall's and König's theorems). Part B is a 10-question quiz (true/false and multiple choice). Attempt every question on your own before consulting Part C (Solutions) at the end of this document.

Part A: Theory Questions

- A1.** Define a *bipartite graph* in two equivalent ways: via a vertex 2-coloring, and via cycle lengths. Briefly explain why a BFS/DFS 2-coloring algorithm (Problem 1) detects non-bipartiteness exactly when it finds an edge between two same-colored vertices.
- A2.** State the Maximum Bipartite Matching problem precisely (input, what a matching is, objective). Then describe the standard reduction to maximum flow: which vertices and edges are added, and what capacities they receive.
- A3.** Prove that, in the matching-to-flow reduction, the maximum matching size equals the maximum s - t flow value. Where is the *integrality theorem* used, and why is it essential?
- A4.** Kuhn's algorithm (Problem 2) repeatedly searches for an *augmenting path*. Define an augmenting path with respect to a matching M , and explain why finding one lets you increase the matching size by exactly one. Why does the algorithm terminate, and how many augmentations occur in the worst case?
- A5.** Define the *neighborhood* $N(S)$ of a set $S \subseteq X$ and a *deficient* set. State **Hall's theorem** precisely.
- A6.** Prove the “easy” (necessity) direction of Hall's theorem: if a matching saturates all of X , then $|N(S)| \geq |S|$ for every $S \subseteq X$.
- A7.** Prove the “hard” (sufficiency) direction of Hall's theorem using max-flow min-cut: if every $S \subseteq X$ satisfies $|N(S)| \geq |S|$, then an X -saturating matching exists. (Hint: show the minimum cut of the matching network equals $\min_S (|X| - |S| + |N(S)|)$.)
- A8.** When no perfect matching of X exists, Problem 4 returns a deficient set S as a *certificate*. Explain why a deficient set is a convincing, easily-checkable proof that no X -saturating matching can exist.
- A9.** Define a *vertex cover*. Prove the weak-duality bound: in *any* graph, $|M| \leq |C|$ for every matching M and vertex cover C .

- A10.** State **König's theorem**. Give a concrete example of a *non-bipartite* graph where maximum matching $<$ minimum vertex cover (so König fails), and report both values.
- A11.** Prove König's theorem constructively: starting from a maximum matching M , build a vertex cover of size $|M|$ using alternating paths from the unmatched X -vertices. (This is exactly the construction implemented in Problem 5.)
- A12.** Explain how Hall's theorem can be derived as a corollary of König's theorem.
- A13.** State **Menger's theorem** (edge version) and explain the reduction (Problem 7) that computes the maximum number of edge-disjoint s - t paths. Why does using capacity-1 edges, together with the integrality theorem, guarantee the paths are edge-disjoint?
- A14.** Describe the *node-splitting* gadget (Problem 8) for computing *vertex-disjoint* paths. Give a small example of a graph in which the number of edge-disjoint s - t paths is strictly greater than the number of vertex-disjoint s - t paths.
- A15.** State the *circulation with demands and lower bounds* problem. Explain the two-step reduction to max flow (Problem 11): how lower bounds are eliminated, and how a super-source/super-sink decide feasibility. What is the role of the condition $\sum_v d_v = 0$?
- A16.** Describe the *project selection / maximum-weight closure* problem and its min-cut reduction (Problem 10). Why are the prerequisite edges given *infinite* capacity, and what does the source side of the minimum cut represent?
- A17.** Both **baseball elimination** (Problem 9) and **image segmentation** are solved with flow/cut. For each, state in one or two sentences what the flow/cut value represents and how the answer is read off. What do these two applications, together with project selection, have in common at the level of the reduction technique?

Part B: Quiz

- B1. (True/False)** A graph is bipartite if and only if it contains no odd-length cycle.
- B2. (Multiple choice)** In the standard reduction of bipartite matching to max flow, the edges between the two sides X and Y are given capacity:
- (a) 0
 - (b) 1
 - (c) ∞
 - (d) the degree of the endpoint in X
- B3. (Short answer)** A bipartite graph has $X = \{x_1, x_2\}$, $Y = \{y_1\}$, with edges x_1y_1 and x_2y_1 . Does X have a perfect matching? If not, give a deficient set S and the values $|S|$ and $|N(S)|$.
- B4. (True/False)** In every graph, the size of a maximum matching equals the size of a minimum vertex cover.
- B5. (Multiple choice)** König's theorem states that maximum matching equals minimum vertex cover in:
- (a) all graphs
 - (b) bipartite graphs
 - (c) planar graphs
 - (d) trees only
- B6. (True/False)** To count *vertex-disjoint* s - t paths, one splits each internal vertex v into $v_{\text{in}} \rightarrow v_{\text{out}}$ with capacity 1.
- B7. (Short answer)** In the project-selection min-cut network, why must the prerequisite edges have infinite capacity?
- B8. (Multiple choice)** In baseball elimination, team x is declared *not eliminated* when the maximum flow:
- (a) is zero
 - (b) saturates every edge *into* the sink t
 - (c) saturates every game edge *out of* the source s
 - (d) equals the number of teams
- B9. (Short answer)** In a circulation with demands, after eliminating an edge's lower bound ℓ on edge (u, v) , how do the demands of u and v change, and how does the edge's capacity change?
- B10. (True/False)** Image segmentation (foreground/background labeling with separation penalties) can be solved by computing a minimum s - t cut.

Part C: Solutions

Part A

- A1.** A graph is **bipartite** if its vertices split into $X \cup Y$ with every edge crossing between X and Y ; equivalently, if the vertices can be 2-colored (say colors 0 and 1) so that adjacent vertices get different colors. A second equivalent characterization: a graph is bipartite iff it has *no odd-length cycle*. The BFS 2-coloring algorithm colors a start vertex 0 and alternates colors across edges; if it ever finds an edge whose endpoints already have the *same* color, the two BFS paths to those endpoints, plus that edge, close an *odd* cycle – so the graph cannot be bipartite. If no such edge is ever found, the coloring is a valid bipartition.
- A2. Input:** a bipartite graph $G = (X \cup Y, E)$. A **matching** $M \subseteq E$ is a set of edges no two of which share an endpoint. The **objective** is to find a matching of maximum size. **Reduction to flow:** add a source s and sink t ; add $s \rightarrow x$ of capacity 1 for each $x \in X$; add $y \rightarrow t$ of capacity 1 for each $y \in Y$; orient each original edge as $x \rightarrow y$ with capacity 1. The maximum s – t flow value equals the maximum matching size.
- A3.** *Matching $\Rightarrow$ flow:* a matching M of size k yields a flow of value k by routing one unit along $s \rightarrow x \rightarrow y \rightarrow t$ for each $(x, y) \in M$; disjointness of M keeps every capacity-1 edge within its capacity. *Flow $\Rightarrow$ matching:* by the **integrality theorem**, since all capacities are integers there is a maximum flow that is *integral*, so each edge carries 0 or 1 unit. The saturated middle edges form a set M ; each x receives at most 1 unit from s , so it sends at most 1 unit onward, i.e. x is in at most one edge of M (likewise each y). So M is a matching with $|M| =$ flow value. Integrality is *essential*: without it a max flow could split a unit fractionally across two middle edges, which would not correspond to any matching.
- A4.** An **augmenting path** (w.r.t. M) is a path that starts and ends at *free* vertices and *alternates* between edges not in M and edges in M . Because the two endpoints are free and the path alternates, it contains one more non-matching edge than matching edge; *flipping* the path (removing its matching edges from M and adding its non-matching edges) yields a valid matching with exactly one more edge. Kuhn’s algorithm tries to find an augmenting path from each X -vertex; each success increases $|M|$ by 1, and since $|M| \leq \min(|X|, |Y|)$ there are at most that many augmentations, after which no augmenting path remains and M is maximum (Berge’s theorem).
- A5.** For $S \subseteq X$, $N(S) = \{y \in Y : y \text{ adjacent to some } x \in S\}$. S is **deficient** if $|N(S)| < |S|$. **Hall’s theorem:** a bipartite graph has a matching saturating all of X if and only if $|N(S)| \geq |S|$ for every $S \subseteq X$ (i.e. no deficient set exists).
- A6.** Suppose a matching M saturates every vertex of X . Fix any $S \subseteq X$. Each $x \in S$ is matched to a distinct partner $\mu(x) \in Y$, and $\mu(x) \in N(S)$ since $x \in S$ is adjacent to it. The map $x \mapsto \mu(x)$ is injective (a matching pairs distinct x ’s with distinct y ’s), so $N(S)$ contains at least $|S|$ distinct vertices: $|N(S)| \geq |S|$.
- A7.** Build the matching network. An X -saturating matching exists iff the maximum flow value is $|X|$, iff (max-flow min-cut) the minimum cut capacity is $|X|$. Take any finite-capacity s – t cut (A, B) , $s \in A$. Let $S = X \cap B$. A finite cut cannot include any infinite-capacity edge; here all edges are finite, but the *cheapest* cut that “abandons” S to the sink side must (i) cut the edges $s \rightarrow x$ for each $x \in X \cap A = X \setminus S$, costing $|X| - |S|$, and (ii) cut the edges $y \rightarrow t$ for every $y \in N(S)$ that it keeps on the source side, costing $|N(S)|$ (it must keep $N(S)$ on the

source side, because each middle edge $x \rightarrow y$ with $x \in S$ would otherwise be a source-to-sink crossing it has to pay for, and putting $N(S)$ on the source side and cutting $y \rightarrow t$ is at least as cheap). Hence the minimum cut equals $\min_{S \subseteq X} (|X| - |S| + |N(S)|)$. This is $< |X|$ exactly when some S has $|N(S)| < |S|$. So if *no* set is deficient, the minimum cut is $|X|$, the max flow is $|X|$, and an X -saturating matching exists.

- A8.** A deficient set S has $|N(S)| < |S|$. Any matching can match the vertices of S *only* to vertices of $N(S)$ (those are their only neighbors), and a matching pairs them injectively; but there are fewer than $|S|$ available partners in $N(S)$, so at least one vertex of S must go unmatched. Hence no matching can saturate X . The certificate is *easy to check*: given S , one computes $N(S)$ in $O(E)$ time and verifies $|N(S)| < |S|$ – no need to re-run any matching algorithm.
- A9.** A **vertex cover** C is a set of vertices such that every edge has at least one endpoint in C . *Weak duality*: let M be any matching and C any vertex cover. Each edge of M must be covered, so it has an endpoint in C . Distinct edges of M share no endpoint, so they require *distinct* cover vertices. Therefore $|C| \geq |M|$. Taking the maximum matching and minimum cover gives $\text{max-matching} \leq \text{min-cover}$.
- A10.** **Kőnig's theorem**: in a *bipartite* graph, maximum matching size = minimum vertex cover size. *Non-bipartite counterexample*: the triangle K_3 on vertices $\{1, 2, 3\}$ with edges $\{12, 23, 31\}$. Its maximum matching has size 1 (any two edges share a vertex), but its minimum vertex cover has size 2 (one vertex covers only two of the three edges). So $1 < 2$, and Kőnig fails for this non-bipartite graph. (This is precisely the example checked in Problem 12.)
- A11.** Let M be a maximum matching. Let $U \subseteq X$ be the unmatched X -vertices. *Mark* every vertex reachable from U by an alternating path (non-matching edge $X \rightarrow Y$, then matching edge $Y \rightarrow X$, repeating); call the marked set Z . Define $C = (X \setminus Z) \cup (Y \cap Z)$. *C covers every edge*: if some edge (x, y) had $x \in X \cap Z$ and $y \in Y \setminus Z$, then either $(x, y) \notin M$ (and we could extend the alternating path from the marked x to mark y – contradiction) or $(x, y) \in M$ (and x could only have been marked via its matching partner y , so y is marked – contradiction). $|C| = |M|$: every vertex of C is matched (an unmatched X -vertex lies in $U \subseteq Z$ so is excluded from $X \setminus Z$; an unmatched Y -vertex in Z would give an augmenting path, contradicting maximality), and no matching edge has both endpoints in C . Hence each matching edge contributes exactly one endpoint to C , giving a bijection $M \leftrightarrow C$ and $|C| = |M|$. Combined with weak duality ($|M| \leq |C|$), C is a minimum cover of size exactly the maximum matching.
- A12.** By Kőnig's theorem, the minimum vertex cover of a bipartite graph has size equal to the maximum matching. X has a perfect matching iff the maximum matching size is $|X|$, iff the minimum vertex cover has size $|X|$. One then shows directly that a vertex cover smaller than $|X|$ exists iff some $S \subseteq X$ is deficient: a cover that omits some $x \in S$ must include all of $N(\{x\})$, and aggregating over a deficient S produces a cover of size $< |X|$; conversely a small cover yields a deficient set. Thus “no perfect matching” $\iff$ “a deficient set exists” – which is Hall's theorem.
- A13.** **Menger's theorem (edge version)**: the maximum number of edge-disjoint s – t paths equals the minimum number of edges whose removal disconnects t from s . *Reduction*: give every edge capacity 1 and compute a max flow. By the integrality theorem the max flow is integral, so each edge carries 0 or 1 unit; the positive-flow edges decompose into s – t paths, and since each edge carries at most one unit, *no edge is shared* – the paths are edge-disjoint. The number of such paths equals the flow value, which by max-flow min-cut equals the minimum edge cut.

- A14. Node-splitting:** replace each internal vertex v by two vertices $v_{\text{in}}, v_{\text{out}}$ joined by an edge of capacity 1; reroute every original edge (u, v) as $u_{\text{out}} \rightarrow v_{\text{in}}$ (capacity 1). The capacity-1 internal edge forces at most one path through v , so a max flow counts *vertex*-disjoint paths. *Example:* vertices s, a, b, m, t with edges $s \rightarrow a, s \rightarrow b, a \rightarrow m, b \rightarrow m, m \rightarrow t$. There are 2 *edge*-disjoint paths ($s a m t$ and $s b m t$ – they share no edge), but only 1 *vertex*-disjoint path, since both must pass through the single internal vertex m . (This is the contrast tested between Problems 7 and 8.)
- A15. Problem:** each vertex v has a demand d_v (> 0 absorb, < 0 supply); each edge has a lower bound ℓ_e and capacity c_e ; we seek a flow $f_e \in [\ell_e, c_e]$ with net flow into each v equal to d_v . *Reduction:* (1) *eliminate lower bounds:* for each edge (u, v) with bound $[\ell, c]$, set its capacity to $c - \ell$ and update $d_u += \ell, d_v -= \ell$ (we have committed to push ℓ units). (2) Add a super-source S and super-sink T : for each v with adjusted $d_v < 0$, add $S \rightarrow v$ of capacity $-d_v$; for each v with adjusted $d_v > 0$, add $v \rightarrow T$ of capacity d_v . A feasible circulation exists iff the max S – T flow saturates all S -edges (equivalently equals the total positive demand). The condition $\sum_v d_v = 0$ (after adjustment, total supply = total demand) is *necessary*: flow conservation forces everything supplied to be absorbed, so unequal totals make feasibility impossible – Problem 11 checks this balance before (or alongside) the flow computation.
- A16. Project selection:** each project v has profit p_v (possibly negative); a prerequisite (p, q) requires that selecting p forces selecting q ; we seek a prerequisite-closed set S maximizing $\sum_{v \in S} p_v$. *Min-cut reduction:* $s \rightarrow v$ with capacity p_v for $p_v > 0$; $v \rightarrow t$ with capacity $-p_v$ for $p_v < 0$; prerequisite edges $p \rightarrow q$ with *infinite* capacity. The infinite capacities ensure no *finite* cut ever separates p (source side) from a prerequisite q (sink side) – so the source side is always a *closed* set, i.e. feasible. The source side of the minimum cut is the optimal project set, and the maximum profit is $(\sum_{p_v > 0} p_v) - (\text{min cut})$.
- A17. Baseball elimination:** the max flow equals the number of remaining games (among the other teams) that can be distributed without any team exceeding x ’s best-possible win total W . Team x is *not eliminated* iff the flow *saturates all game edges* (every remaining game can be “played” in the model); otherwise it is eliminated, and the minimum cut names a set of teams that collectively must accumulate too many wins. **Image segmentation:** the minimum s – t cut’s capacity equals the total cost $\sum_{i \in F} b_i + \sum_{j \in B} a_j + \sum_{\text{split}} p_{ij}$, and the source/sink sides of the cut give the foreground/background labeling. *Common technique:* all three (baseball, project selection, segmentation) are **min-cut** reductions – they encode a combinatorial trade-off as the capacity of a cut, then read the optimal partition off the source side. (Baseball is most naturally posed as a *saturation* check, which is the max-flow side of the same coin.)

Part B

- B1. True.** A graph is bipartite iff it can be 2-colored iff it has no odd cycle – these are three equivalent characterizations.
- B2. (b) 1.** The middle edges $x \rightarrow y$ get capacity 1; the capacity-1 edges from s and into t then enforce that each vertex is matched at most once. (Using capacity ∞ on middle edges also works since the endpoint edges already cap usage, but capacity 1 is the standard choice.)
- B3. No,** X has no perfect matching. Both x_1 and x_2 have their only neighbor y_1 , so $S = \{x_1, x_2\}$ is deficient: $|S| = 2$ but $|N(S)| = |\{y_1\}| = 1 < 2$.

- B4. False.** This equality (König’s theorem) holds only for bipartite graphs. In general, $\text{max-matching} \leq \text{min-cover}$, and the triangle K_3 shows the inequality can be strict ($1 < 2$).
- B5. (b) bipartite graphs.** König’s theorem is specifically about bipartite graphs; the analogous statement fails for general graphs.
- B6. True.** The internal edge $v_{\text{in}} \rightarrow v_{\text{out}}$ of capacity 1 limits the flow through v to one unit, forcing the paths to be vertex-disjoint.
- B7.** Infinite capacity guarantees that *no finite-capacity cut* ever cuts a prerequisite edge, so a project p on the source side can never be separated from its prerequisite q . This forces the source side of any minimum (finite) cut to be *closed* under prerequisites – i.e. a feasible project set.
- B8. (c) saturates every game edge out of the source s .** If all remaining inter-team games can be routed to winners without pushing any team past $W = w_x + (\text{remaining}_x)$, then x can still finish (tied for) first.
- B9.** The capacity becomes $c - \ell$ (we “pre-pushed” ℓ units), the demand of u increases by ℓ ($d_u += \ell$, since u must now also account for sending those ℓ units), and the demand of v decreases by ℓ ($d_v -= \ell$, since v already receives those ℓ units).
- B10. True.** Foreground/background labeling with per-pixel likelihoods and pairwise separation penalties is exactly a minimum s – t cut: s is the foreground terminal, t the background terminal, and the cut capacity equals the total cost being minimized.